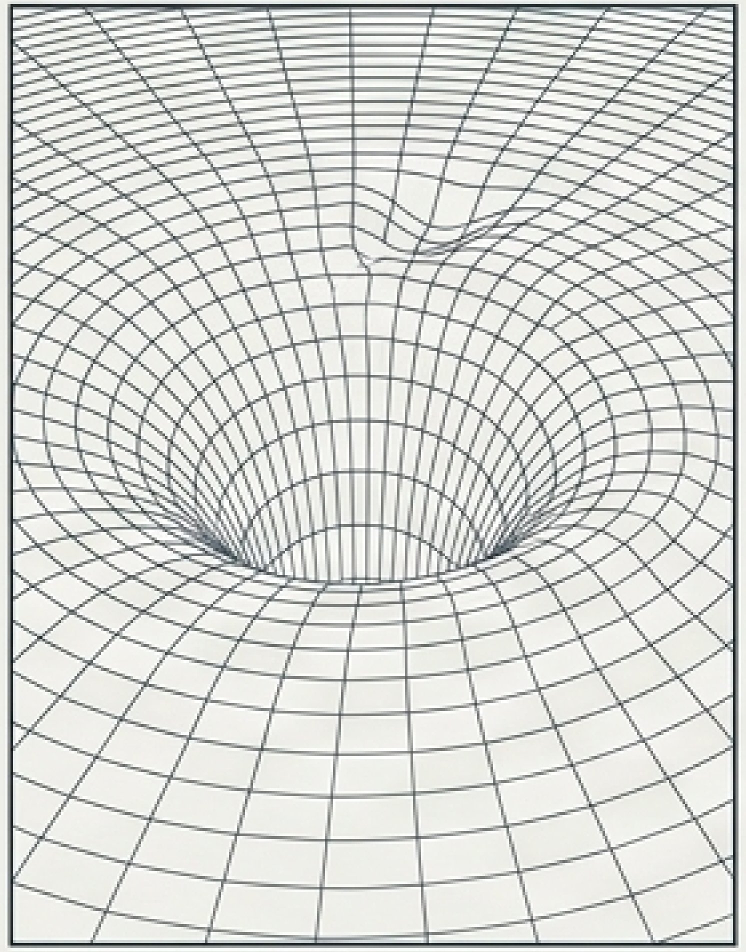
Compute-Based Reality & The Mass Gap

Could a computational foundation lead to a universal theory of physics?

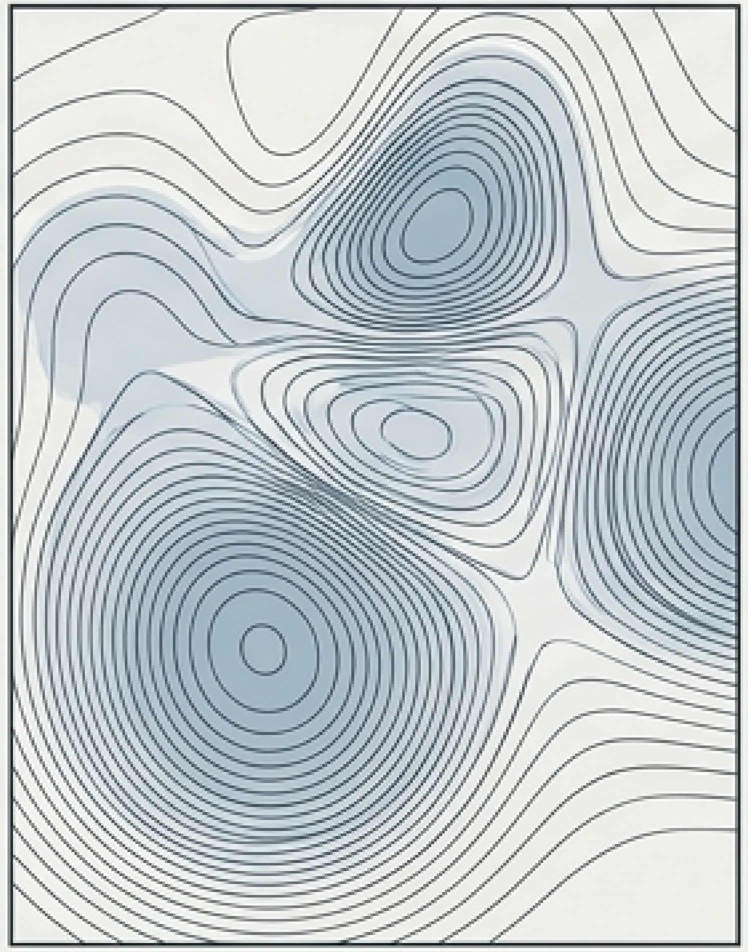
Physics has highly successful theories, but lacks a coherent foundation.

> We know how they work. We don't know why they have these specific mathematical structures.

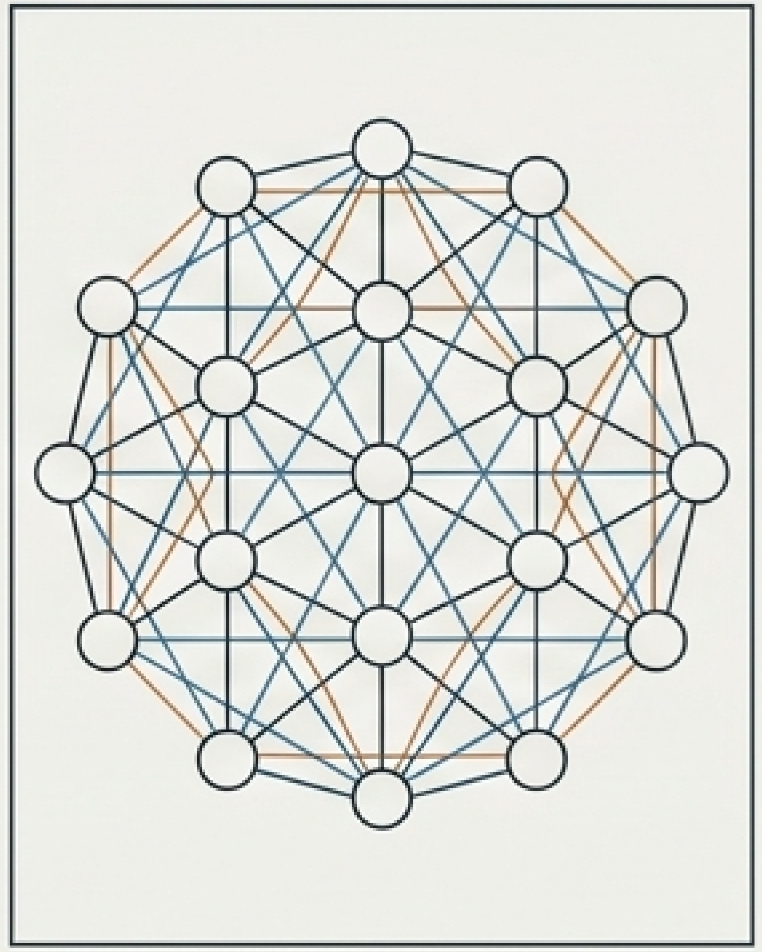
General Relativity



Quantum Field Theory



The Standard Model



The Unsolved Test:

The Yang-Mills Existence & Mass Gap

A rigorous construction of a non-trivial quantum Yang-Mills theory on R^4 with a strictly positive mass gap.

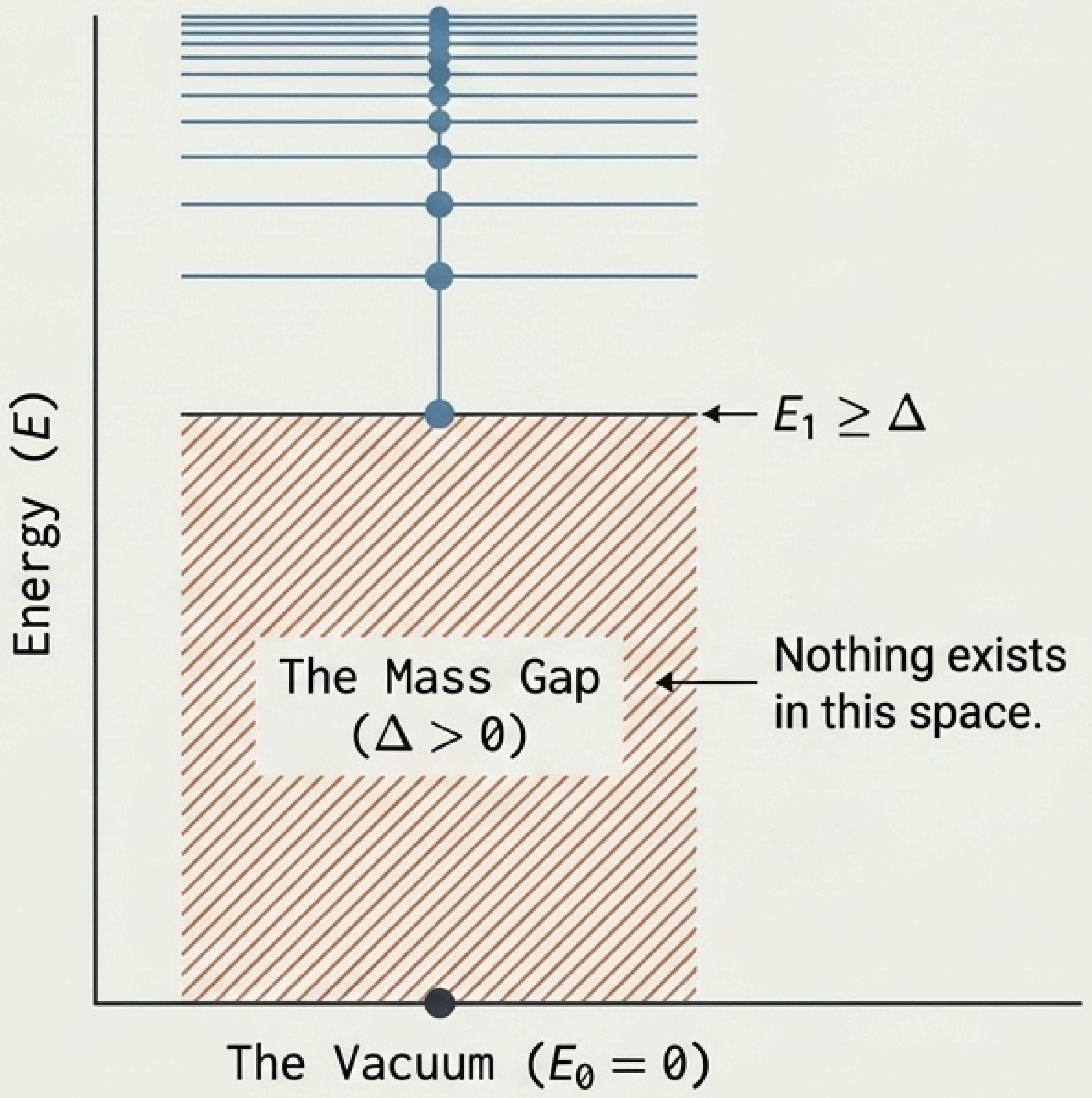
`Requirement: \Δ > 0`

The Conceptual Paradox of the Vacuum

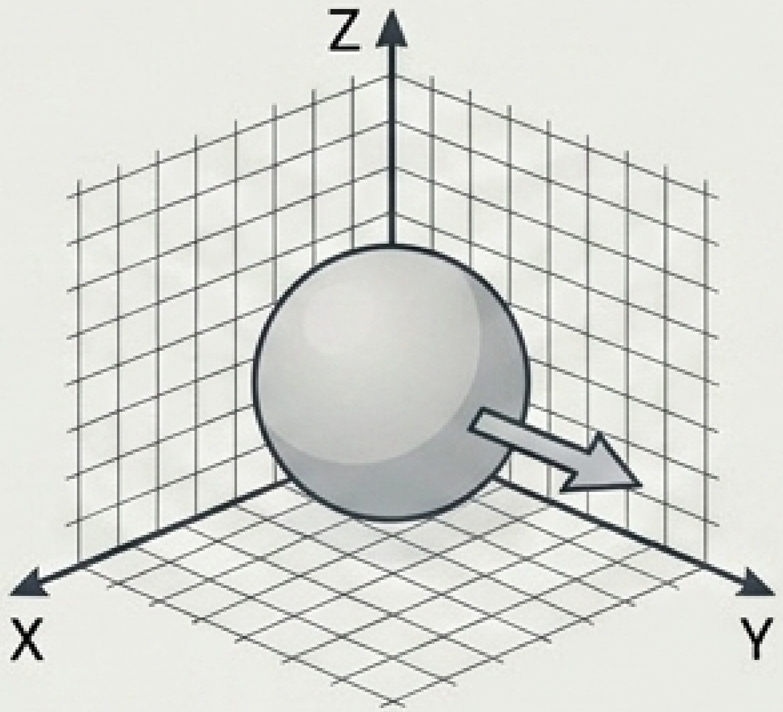
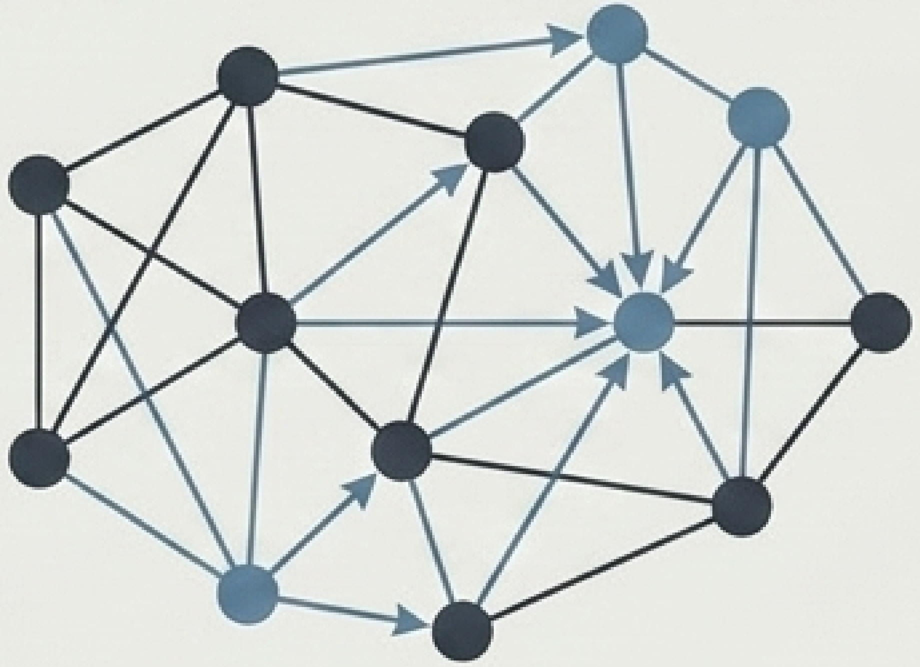
Observation: Classical Yang-Mills equations contain completely massless fields.

The Paradox: Quantum theory demands a finite positive energy gap. There is no sequence of physical excitations whose energies continuously approach zero.

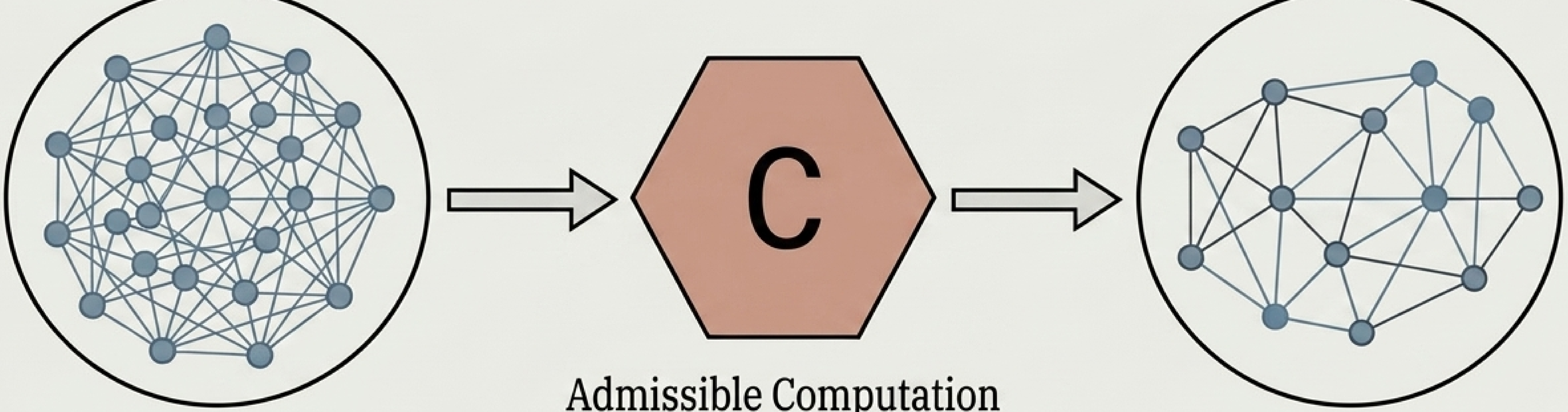
What property of reality prevents provictional arbitrarily small physical excitations from existing?



A new primitive ontology for reality.

	Conventional Assumption	CBR Assumption
The Substrate	A pre-existing container (Space/Time)	A self-consistent computational network
The Ontology	Objects + Space + Time + Laws	State + Relations + Allowed Transformations
Visual Representation		
The Action	Objects evolving through the container	The continuing resolution of relationships

Reality as an ongoing constraint-satisfaction process.



Admissible Computation

R_t
Relational State

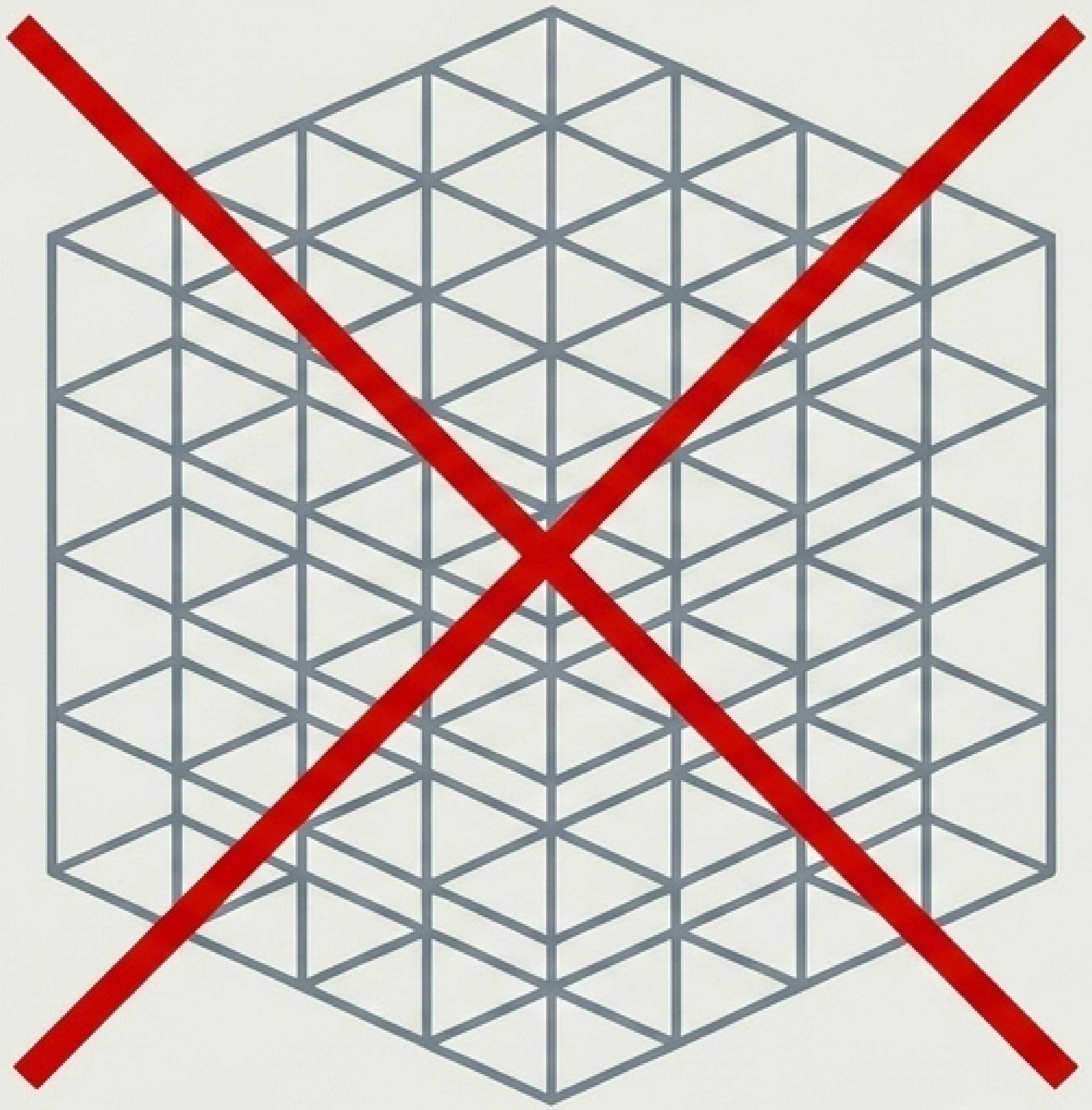
$$R_{t+1} = C(R_t)$$

R_{t+1}
Next Consistent State

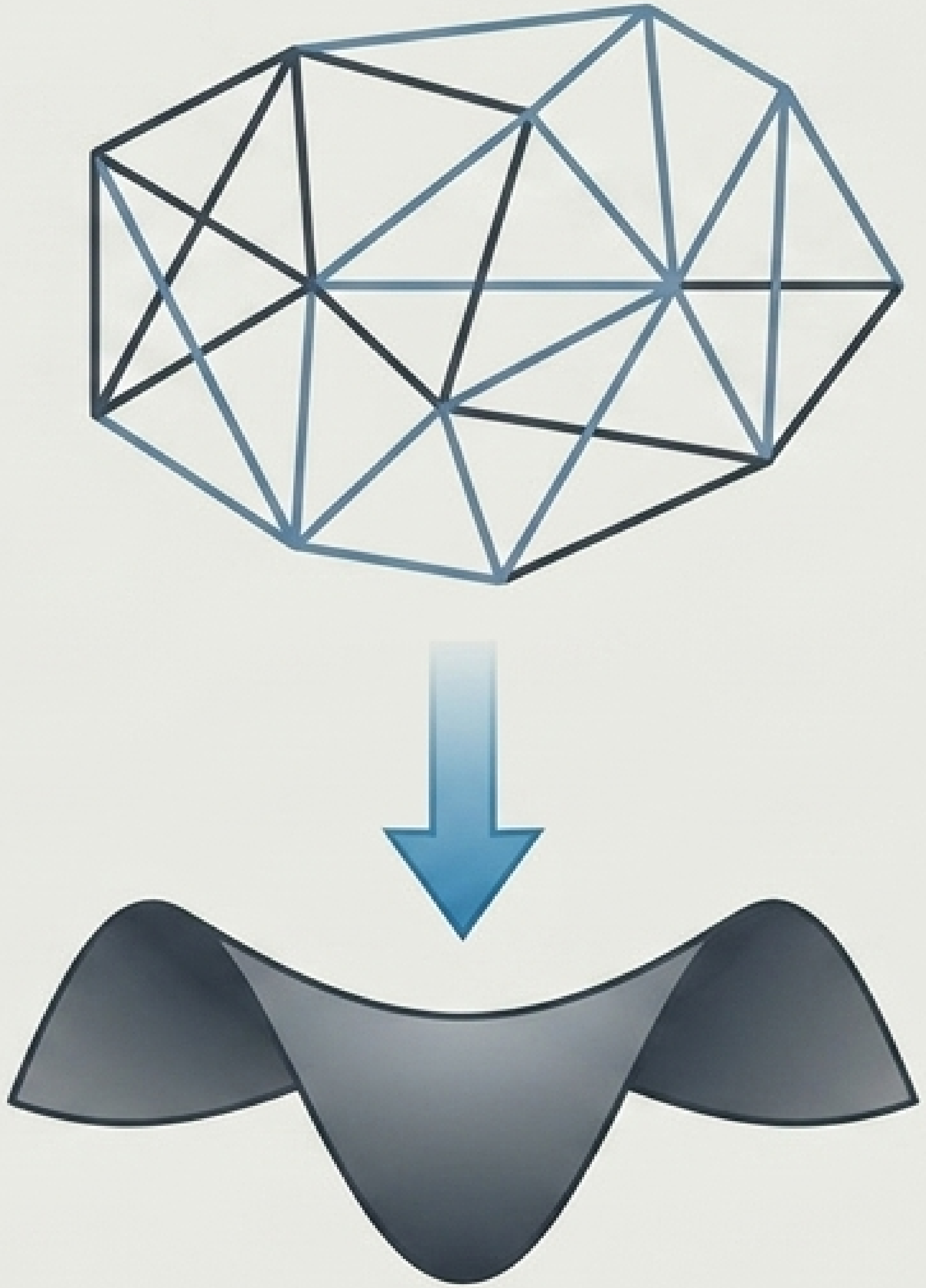
Reality cannot arbitrarily transition between states. Only transformations satisfying a consistency operator ($\Phi(G) = 0$) are physically admissible.

Crucial Distinction: CBR is not a fixed static lattice.

Myth: Reality is a grid.



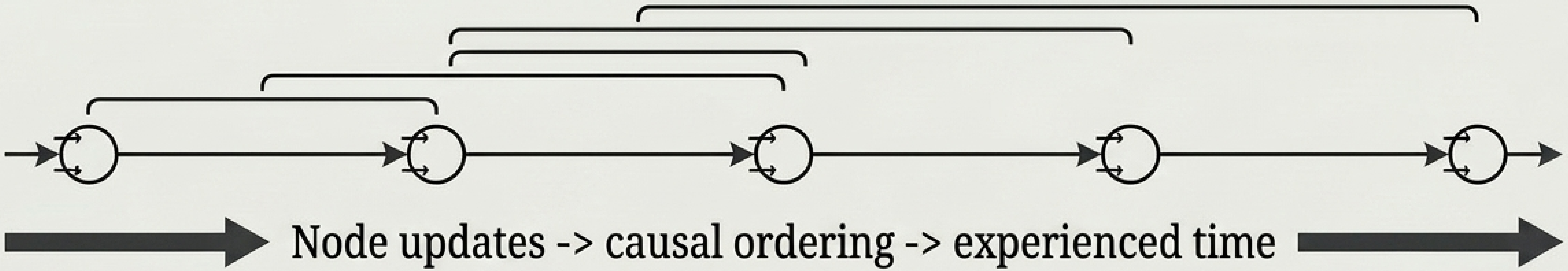
Fact: Reality is an evolving complex.

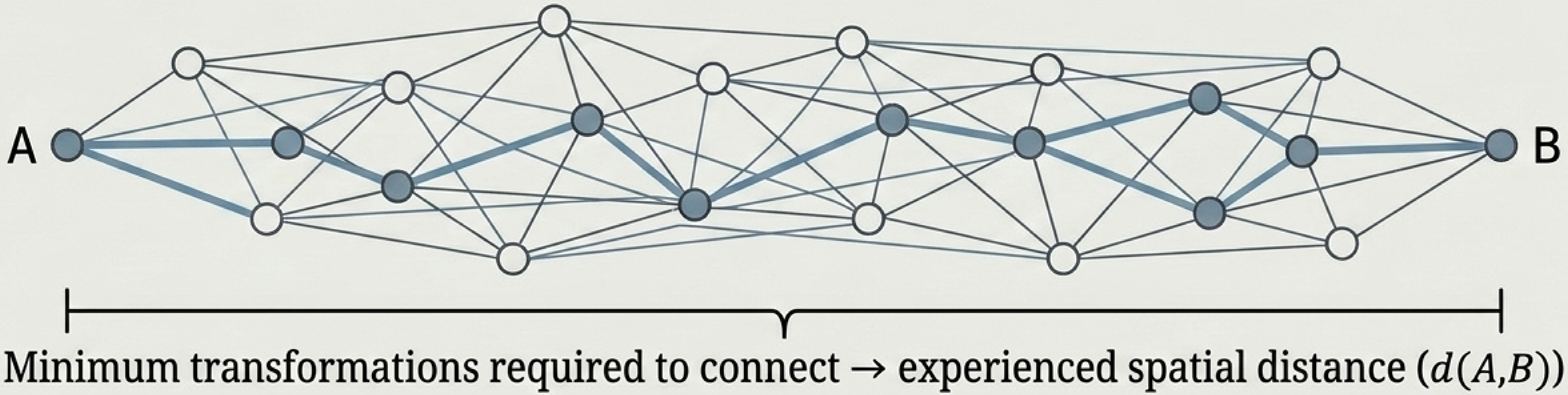


CBR is relational, dynamical, and background-independent.

The effective large-scale geometry of the continuum emerges only after macroscopic coarse-graining.

Space and Time are not the substrate. They are properties of computation.



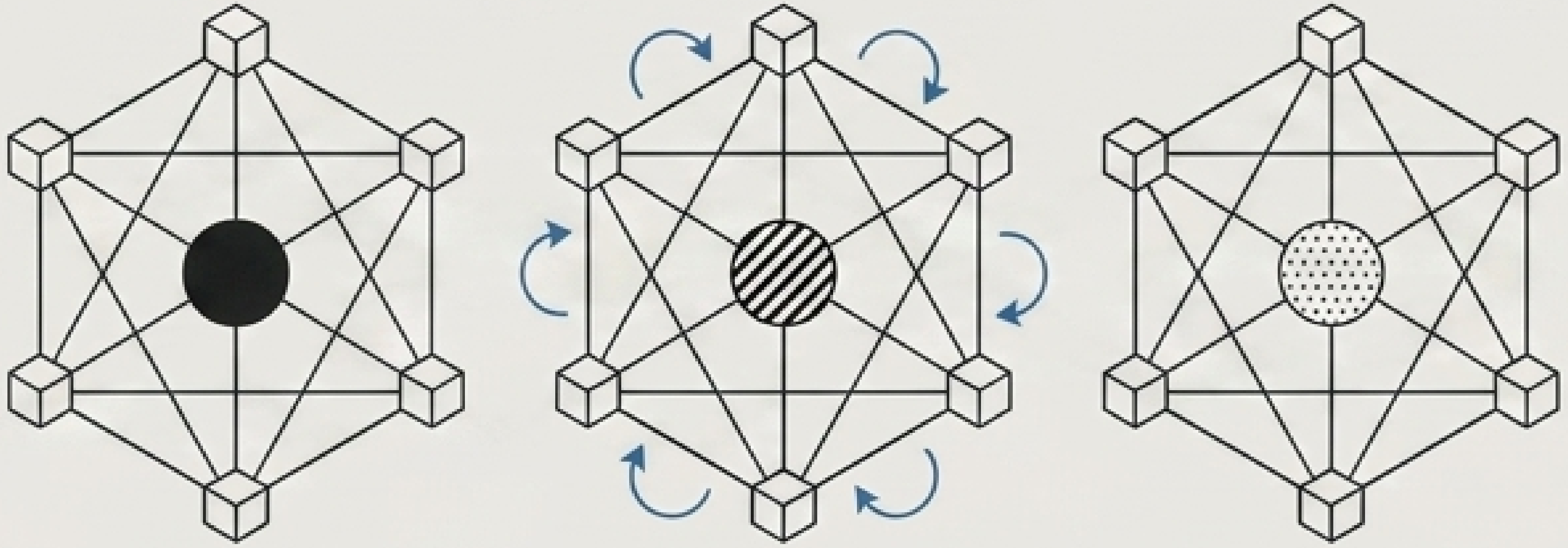


Time is the ordering induced by successful state transformations.
Spatial separation is computational distance.

Redefining Gauge Symmetry and Field Curvature

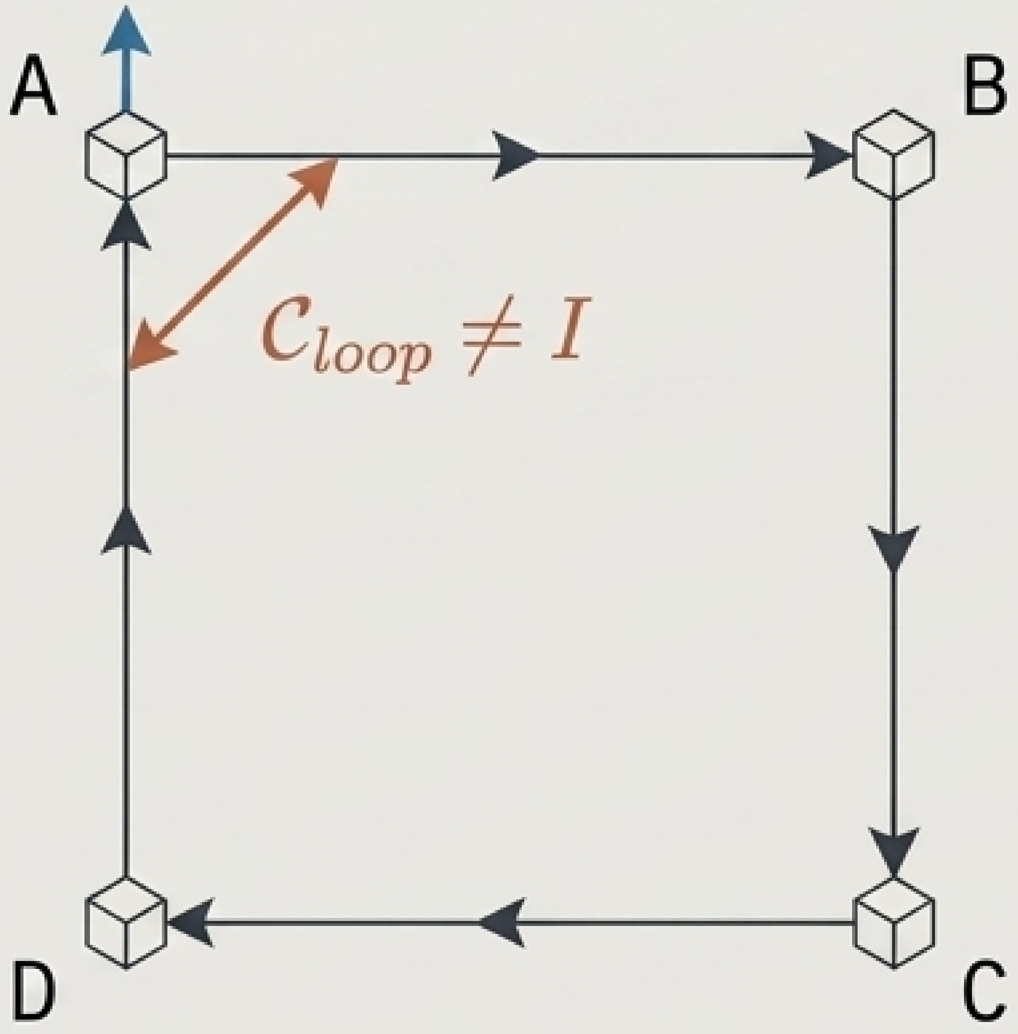
Gauge Redundancy

Physics lives in relational invariants, not absolute representations. Gauge freedom is computational redundancy ($\psi_x \to g_x\psi_x$).



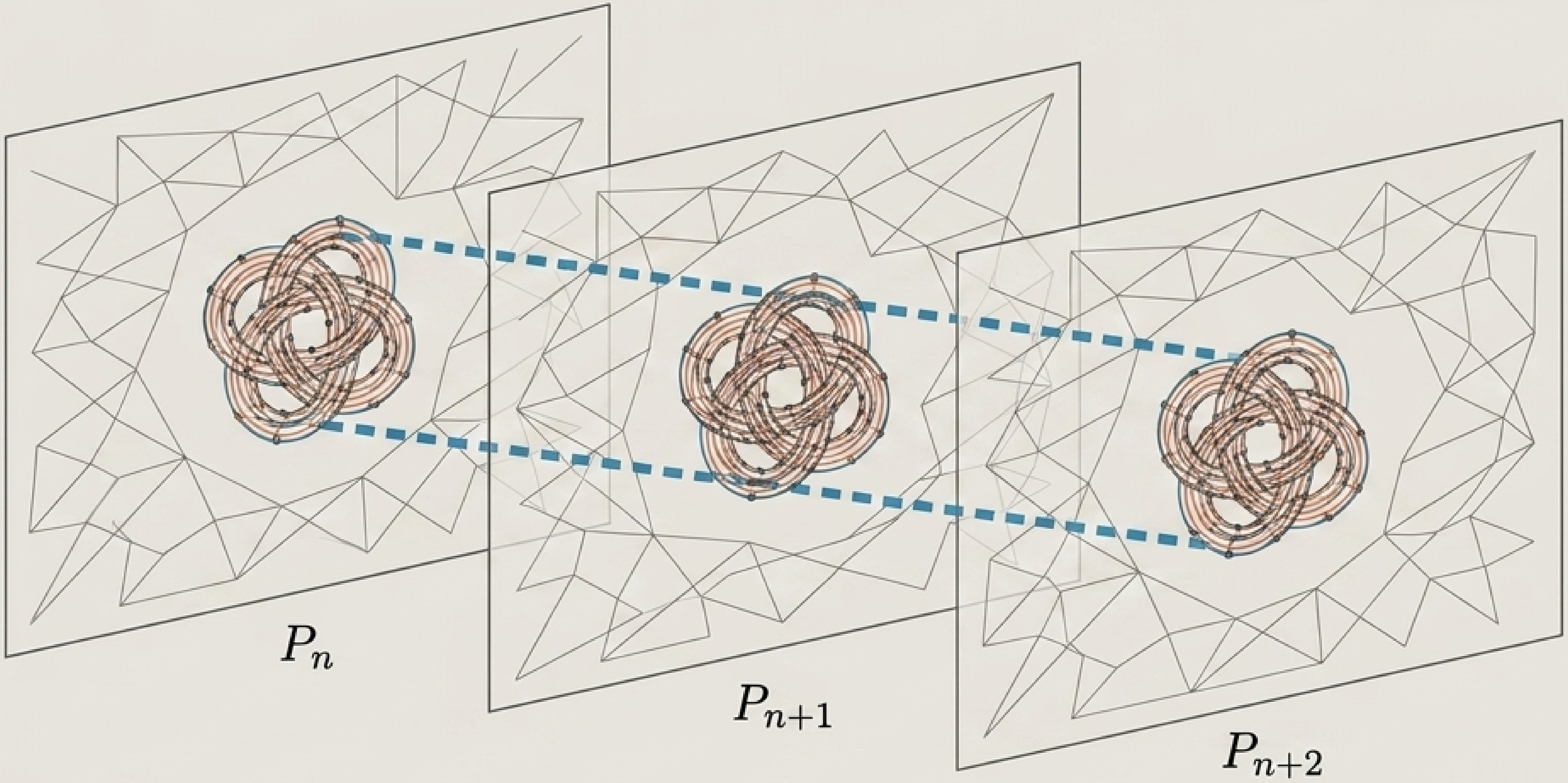
Curvature

Curvature ($F_{\mu\nu}$) measures the failure of relational computation to compose trivially around a loop. Field strength is computational holonomy.



Matter as a Topological Knot

A particle is not a tiny object occupying space. It is a stable, self-maintaining pattern in relational computation.

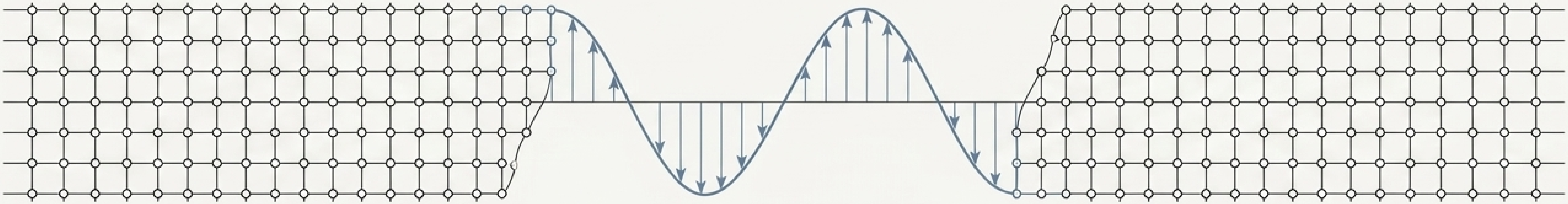


Particle stability $\approx$ topological/computational obstruction.

$$I(P_n) = I(P_{n+1})$$

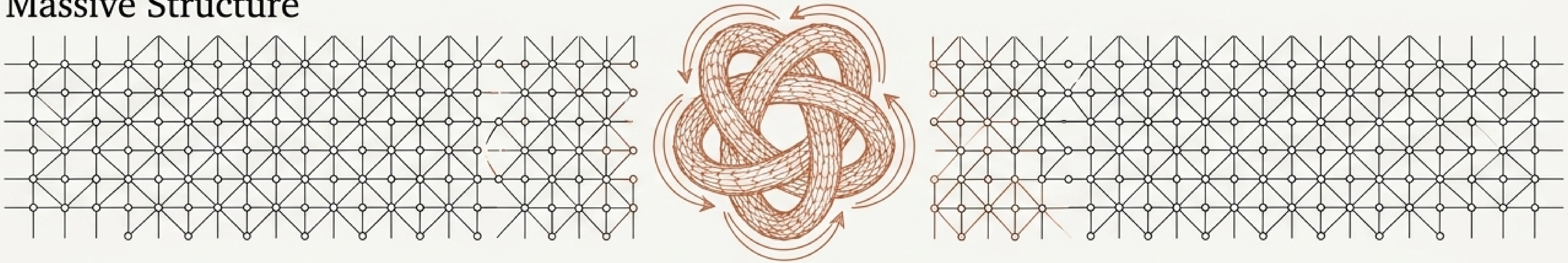
Mass is the computational cost of persistence

Massless Excitation



Zero persistent internal structural updates

Massive Structure



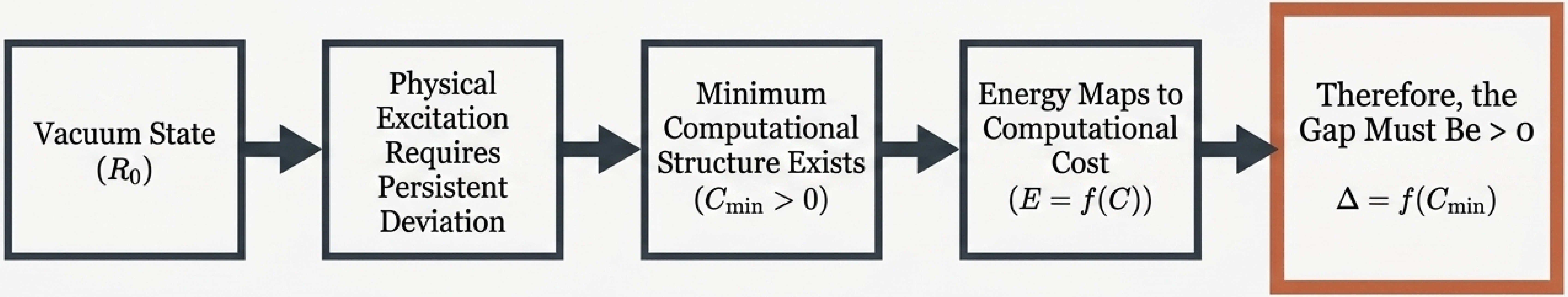
High recurring update cost

$$\boxed{m \propto \text{irreducible update cost of maintaining a persistent excitation.}}$$

Mass measures the computational resources required for a structure to persist while reality updates.

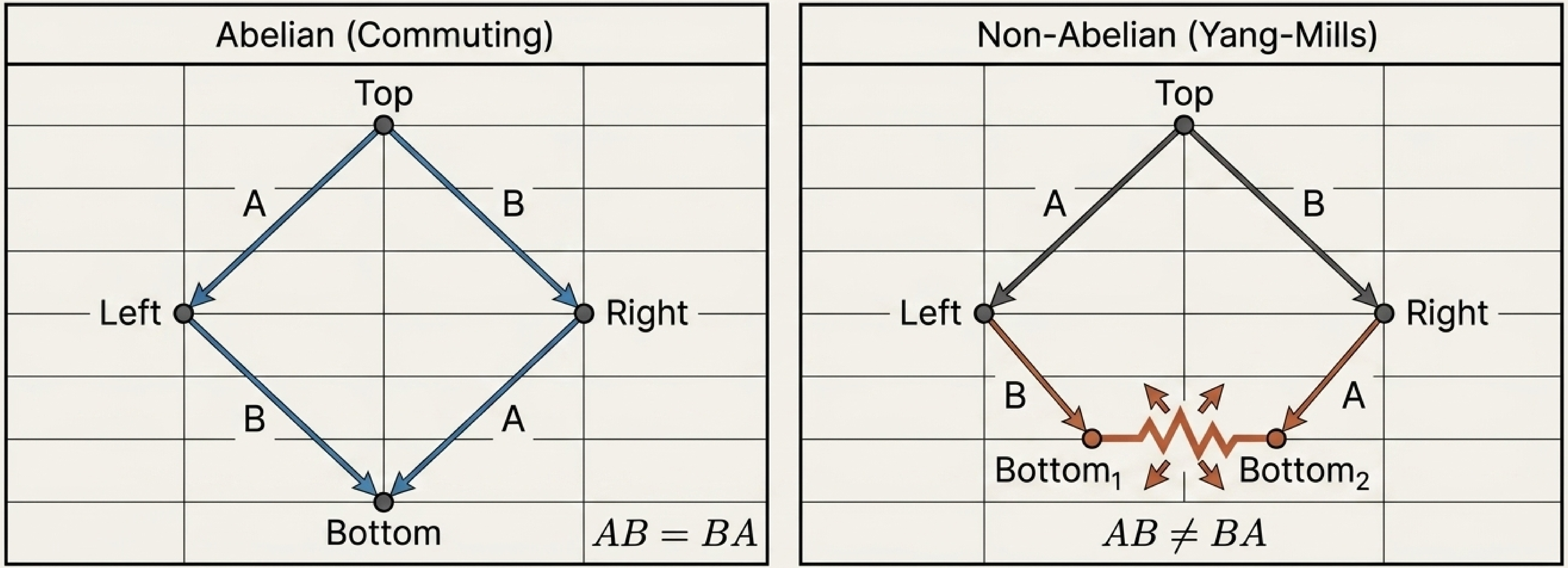
The CBR Interpretation of the Mass Gap

The mass gap doesn't need to be mathematically inserted into continuum fields. It emerges naturally from computational irreducibility. There is no arbitrarily small non-trivial persistent computation.



Where Yang-Mills meets Compute: Order matters.

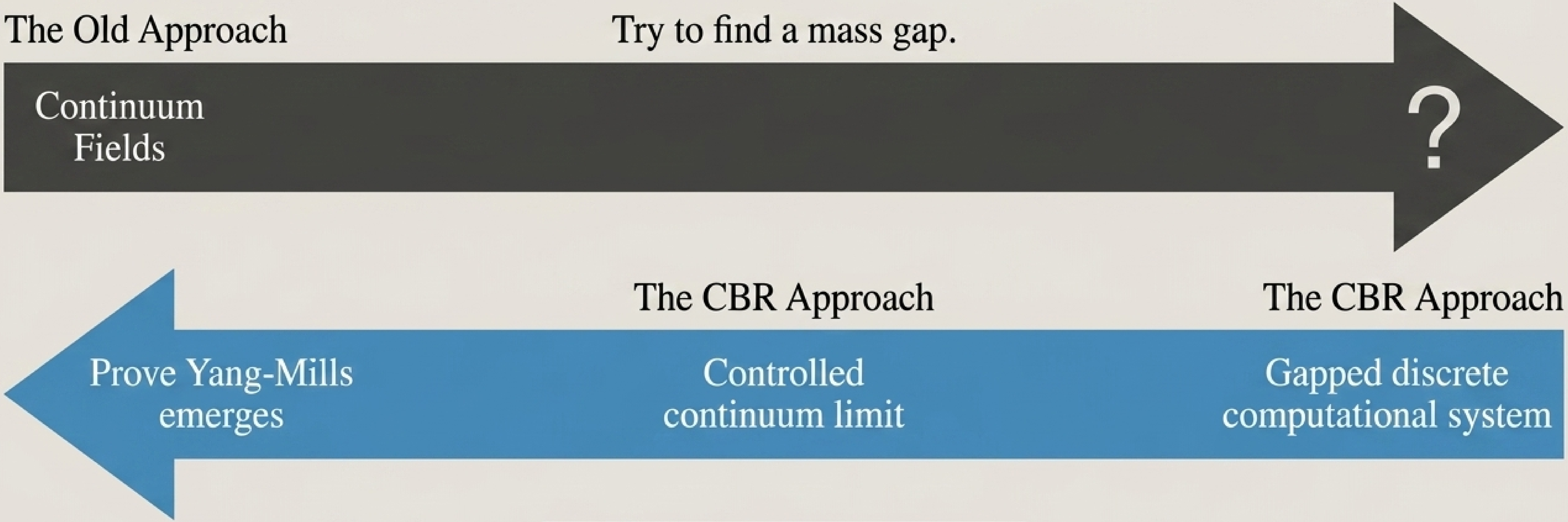
Non-commuting computation → Irreducible relational curvature → Minimum stable excitation → Mass Gap



Frustration Functional: A non-Abelian network possesses loops whose local requirements cannot all be simultaneously trivialized ($F(G) > 0$).

Inverting the Problem

Rather than starting with continuum fields and hoping g a mass gap appears, begin with a system where a gap is computationally unavoidable, then prove Yang-Mills emerges as its continuum description.



$$\boxed{\lim_{a\to 0} \Delta(a) = \Delta_{\mathrm{YM}} > 0}$$

The 7 Axioms of Compute-Based Reality

The Goal: If Axiom 7 can be proven as a natural consequence of Axioms 1-6 (locality + gauge invariance + global consistency + non-Abelian structure), the Mass Gap is mathematically solved.

1. Relationality

2. Local computability

3. Finite transformation

4. Representation invariance (The origin of gauge symmetry)

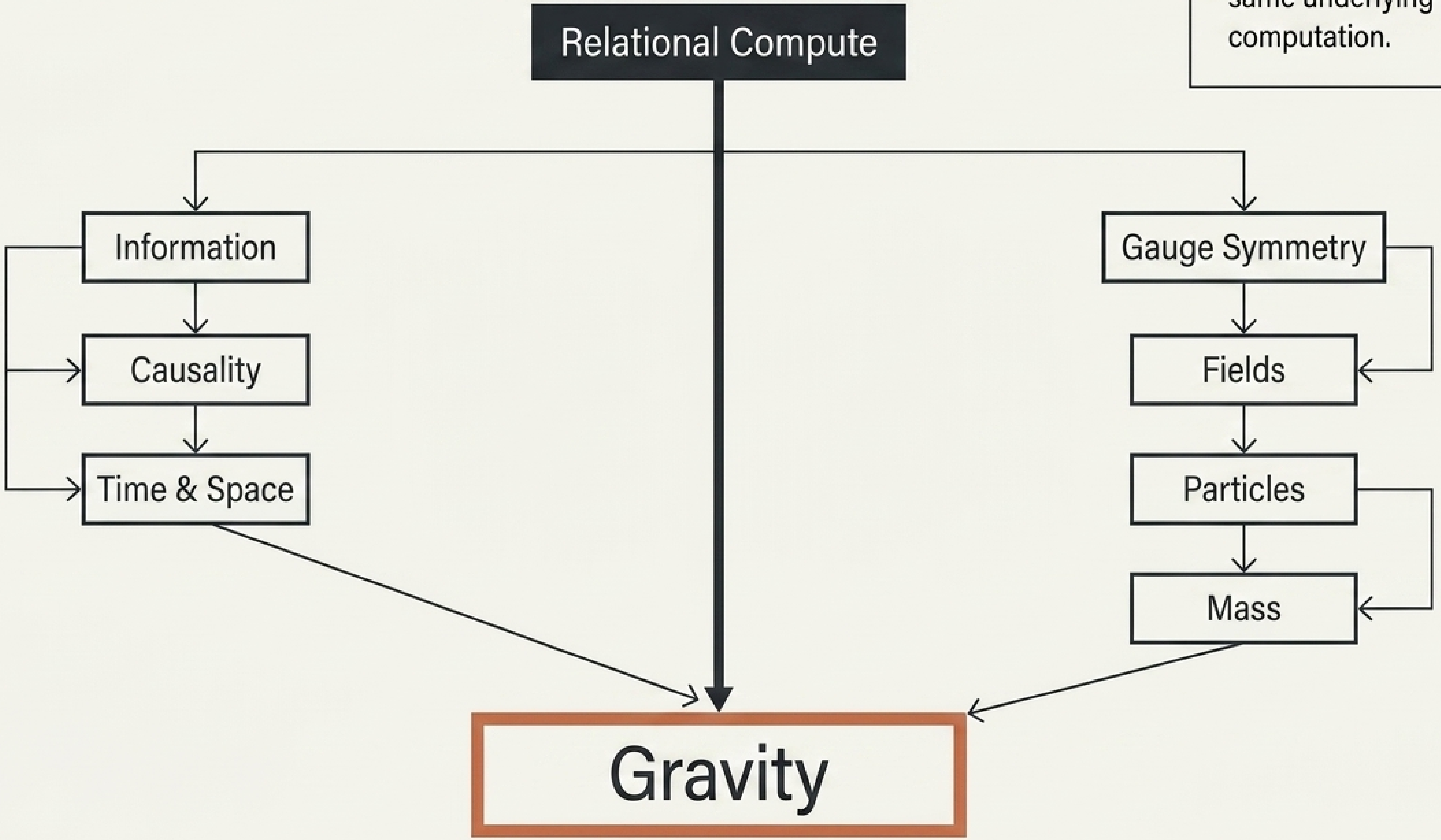
5. Global consistency

6. Persistent distinguishability

7. Irreducibility (There exists a minimum non-trivial persistent transformation)

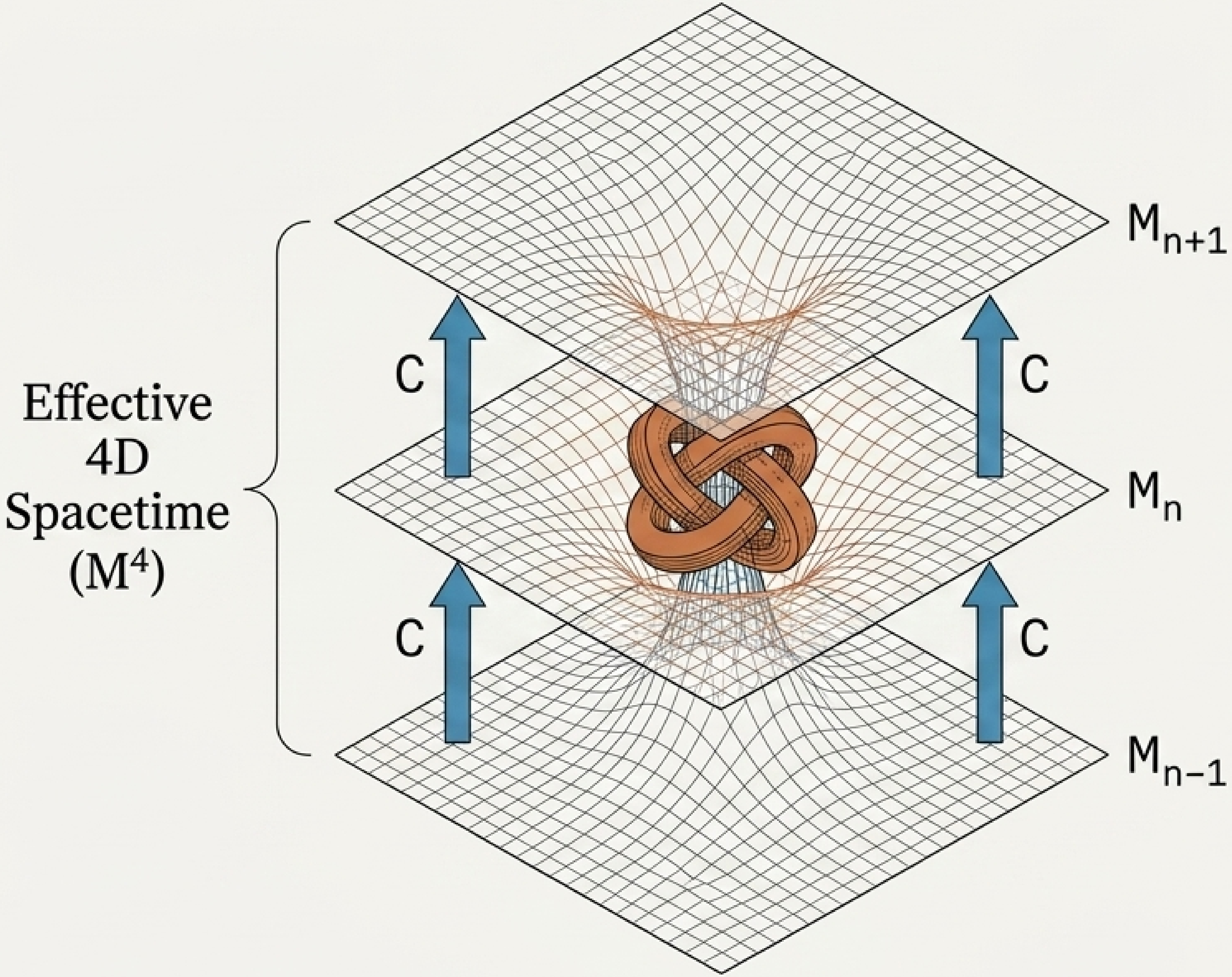
A universal hierarchy of emergence.

Quantum fields and spacetime geometry may simply be two macroscopic descriptions of the same underlying `relational` computation.



Gravity is the geometry induced by computation.

3-manifold + state
transition s ate transition
→ 4-dimensional spacetime.

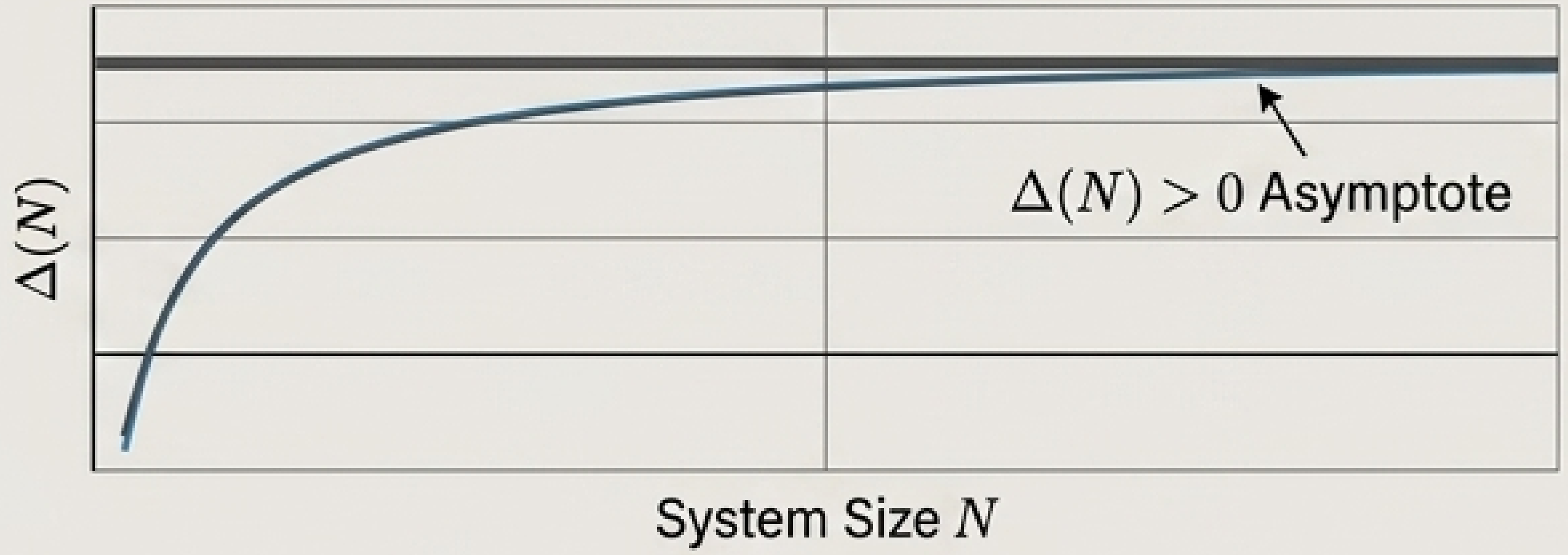


Computation does not happen inside spacetime. Its repeated action generates spacetime. Matter consumes computational capacity; gravity is the resulting warped geometry of what remains accessible.

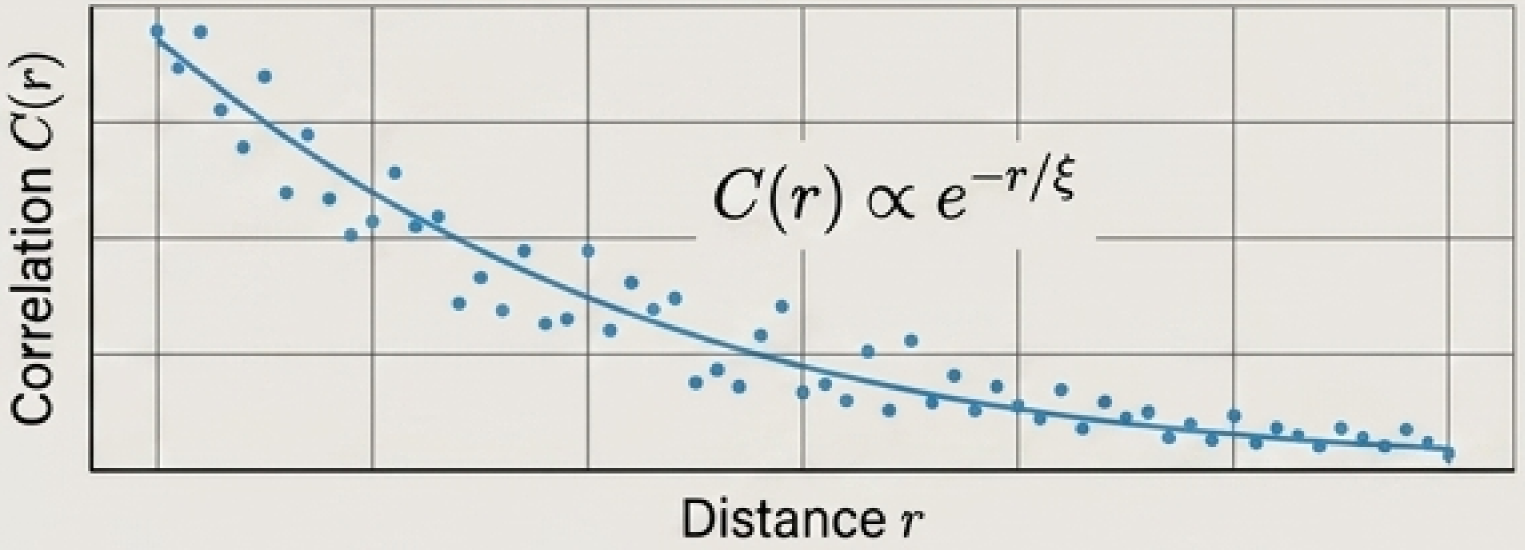
Simulation Before Proof.

Numerical results in finite relational models will not prove the Millennium conjecture. But searching computationally for models exhibiting a positive spectral gap will identify the exact mathematical structures worth rigorously proving.

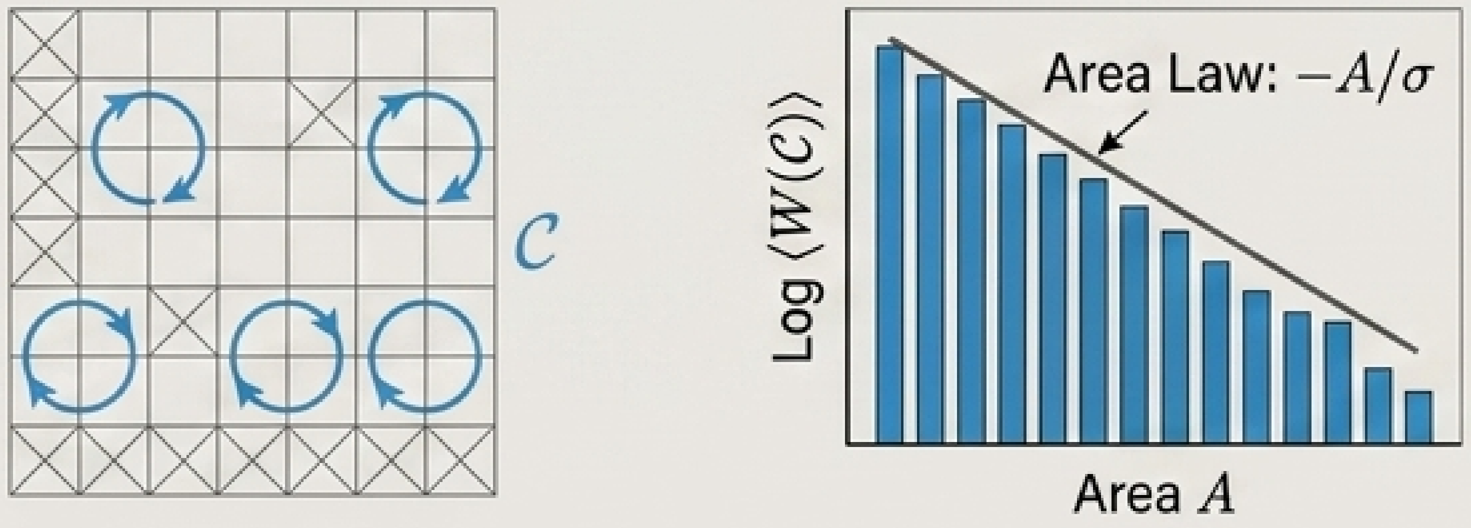
Spectral Gap ($\Delta(N) > 0$)



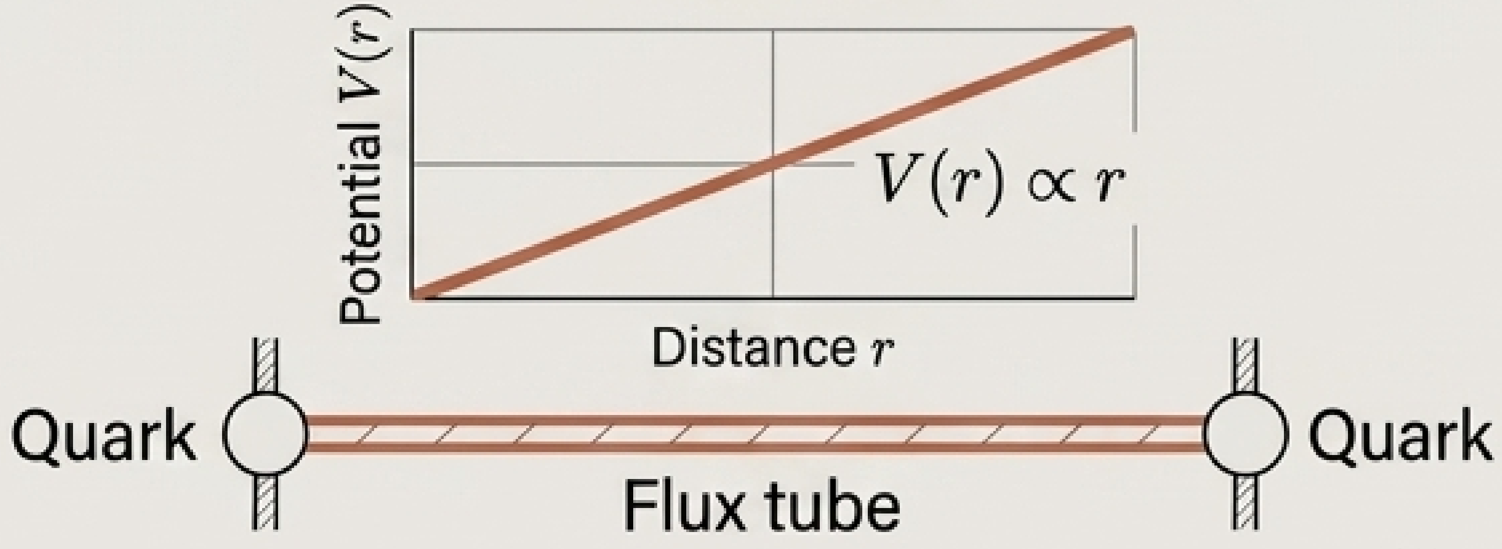
Correlation Length



Wilson Loops



Confinement



The Falsification Programme: CBR becomes physics only when it can fail.

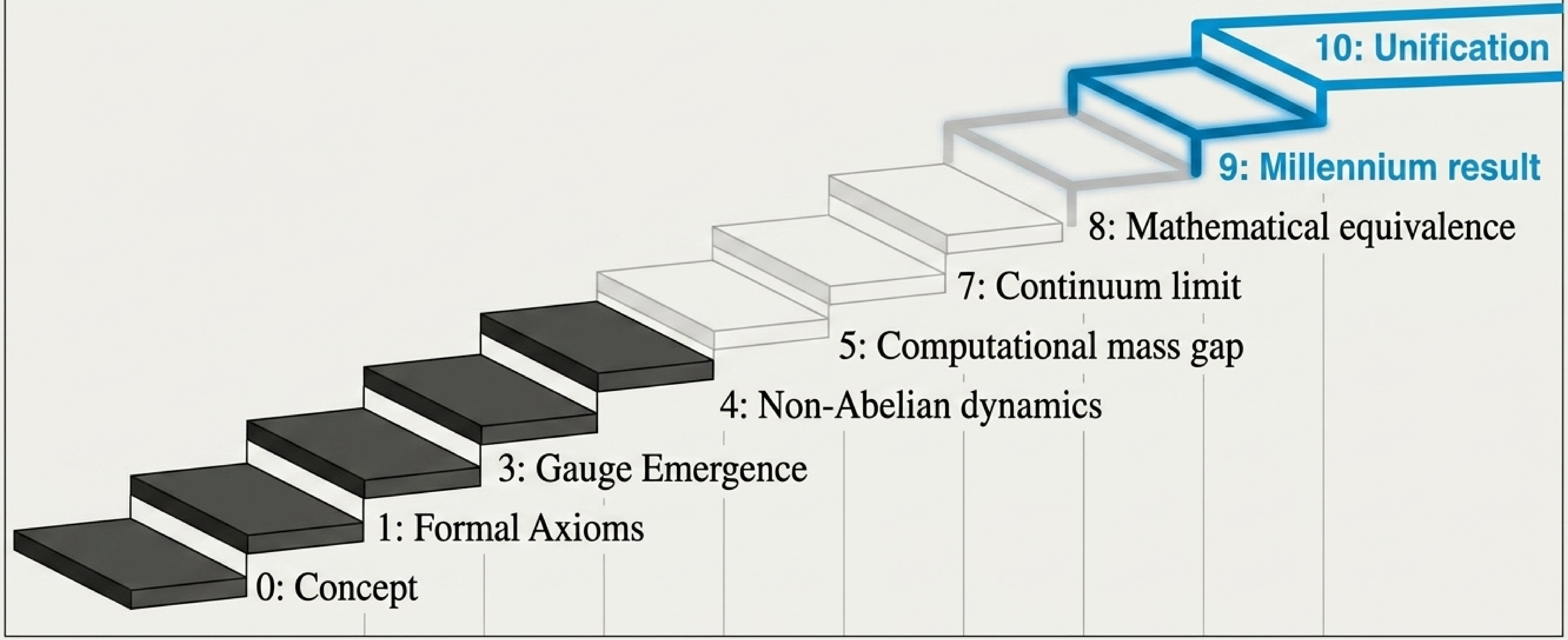
True scientific power comes from prediction. A compelling result requires that the numerous parameters of the Standard Model emerge naturally from a much smaller computational parameter set.

- ☐ 1. Dimensionality of macroscopic spacetime
- ☐ 2. Lorentz invariance
- ☐ 3. Allowed gauge groups
- ☐ 4. Particle mass ratios
- ☐ 5. Coupling constants
- ☐ 6. Cosmological constant
- ☐ 7. Black hole entropy
- ☐ 8. Quantum entanglement structure
- ☐ 9. Gravitational behaviour
- ☐ 10. The Yang-Mills mass gap

$$\{\text{CBR parameters}\} \ll \{\text{Standard Model parameters}\}$$

The 10 Steps to a Universal Theory.

We are moving from formal axioms toward proving $\Delta_{\mathrm{YM}} > 0$. The progression is **strictly sequential; you cannot skip steps to reach unification.**



The Central Conjecture.

Theorem 1: CBR Mass Gap Conjecture

Any non-trivial, locally finite, globally consistent, non-Abelian relational computational system possesses a strictly positive minimum excitation cost.

$$\inf_{X \neq 0} C(X) > 0$$

Theorem 2: CBR-YM Correspondence

There exists a CBR system whose continuum limit is 4D Yang-Mills theory, where the computational gap maps perfectly to the YM mass gap.

Conclusion: $\Delta_{CBR} > 0 \Rightarrow \Delta_{YM} > 0$

Necessity, not Contingency.

Traditional physics asks: What are the things from which the universe is made? CBR asks: What relationships can persist consistently?

Do not assume the laws of physics. Derive why computation must look like physics. A universal theory explains why reality has the space of possibilities that it does.